सत्यमेव जयते

भारत सरकार/**Government of India**
विद्युत मंत्रालय/**Ministry of Power**
केंद्रीय विद्युत प्राधिकरण/**Central Electricity Authority**
जल विद्युत अभियांत्रिकी व प्रौद्योगिकी विकास एवं नवीनीकरण व आधुनिकीकरण प्रभाग
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No. 10/3/HE&TD/2024/ Date: .01.2024

Sh. BalRam Bhardwaj
Innovative MD, M/s MACLEC,
Matiyala Industrial Area,
Delhi-110059

Subject: CEA Report of the Committee "To Study The Concept & Commercial Applications of Hydro Kinetic Turbine Developed By M/S Maclec" - Reg.

Sir,

This is with reference to your email dated 27.10.2023 on the subject vide which it was requested to revisit the CEA Report based on the present technology readiness level, market acceptance, impact created, orders in hand as well as global reputation.

It is to mention in this regard that the CEA Report "To study the Concept and Commercial applications of Hydro Kinetic Turbines (HKTs) developed by M/s Maclec Pvt. Ltd. Delhi" has already been submitted to Ministry of Power (MoP) in October'2021, **which is comprehensive & complete and already addresses all the queries raised by M/s MACLEC**. Subsequently, the subject matter was referred to MNRE, as MoP is mandated for the Hydro Electric Projects with installed capacity of 25MW & above.

Now, M/s MACLEC has raised some queries vis-à-vis the submitted CEA Report. The subject matter has been examined and we have following observations on the technical aspects:

1. "**Impact on Canals/ Rivers/ Streams"** and **"Replicability of technology in Running Streams"**

The submitted CEA Report clearly mentions that the HKTs may be installed in canals, rivulets, upstream/ downstream channels of existing/ upcoming conventional HPPs, cooling water channel of thermal power plants, urban waste water/ sewage canals, tidal rivers and oceans etc. However, the feasibility of installation of HKTs may vary as per the site conditions, soil strength, sufficient water velocity & depth of river bed, wake recovery distance in case of HKT farm/ array, use of water channels for transportation/ recreational purposes etc. For these aspects, the study may be carried out by M/s MACLEC as required for different locations/ sites/ rivers/ canals/ streams.

2. **Technology Readiness Level (TRL)**

It may be mentioned that TRL of 6 to 7 in the year 2021 for HKTs was indicated in the report based on the data provided by M/s MACLEC and deliberations held with committee members. However, technology upgradation, changes in data and new implementation of HKTs being a continuous process, TRL may change. Accordingly, the progress report of 3x5kW HKTs in Ramnagar, Uttarakhand has been perused and it is observed that these HKT units have been operating over the full range of expected conditions for the past 2 years. As such, the TRL of technology has certainly upgraded and on the basis of data provided by M/s MACLEC, the TRL may be considered at level 9.

3. **Royalty charges for water**
Ministry of Power vide order dated 25.04.2023 has termed imposition of taxes/ duties including water cess on generation of electricity as illegal and unconstitutional.

4. **Allotment of Projects on nomination basis**
Water and Water Power is State subject and allotment of Hydro power projects for development is being done by the State Governments. Further, it is opined that the state governments may adopt transparent and objective criteria for allotment of sites for installation of SHK turbines to ensure fair play and attract competition in the sector so that precious natural resources are utilized in the most economical and efficient manner.

Yours faithfully,

(Reetesh Tiwari)
Dy. Director

Copy to:
1. Chief Engineer (HPP&I/ F&CA), CEA